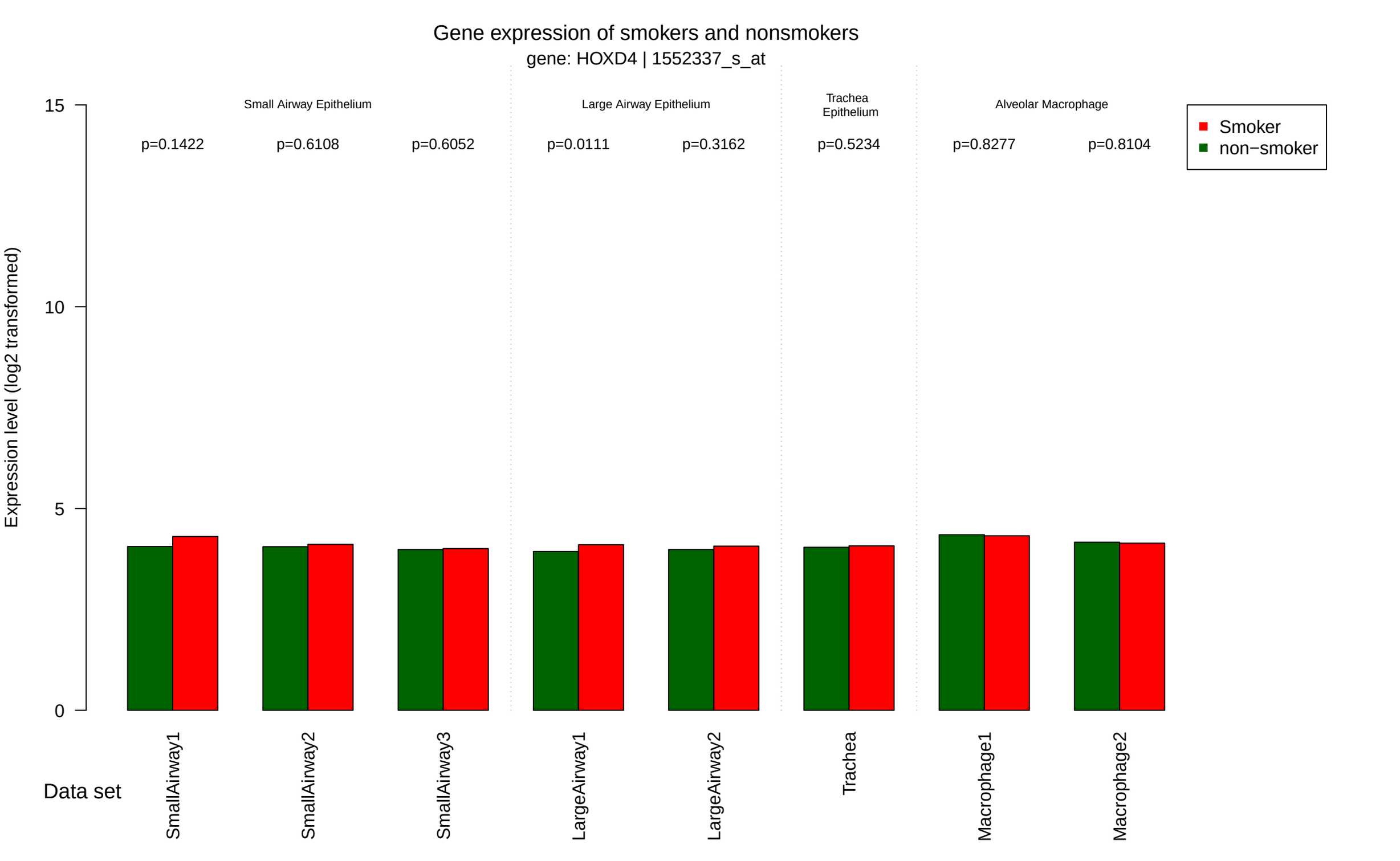
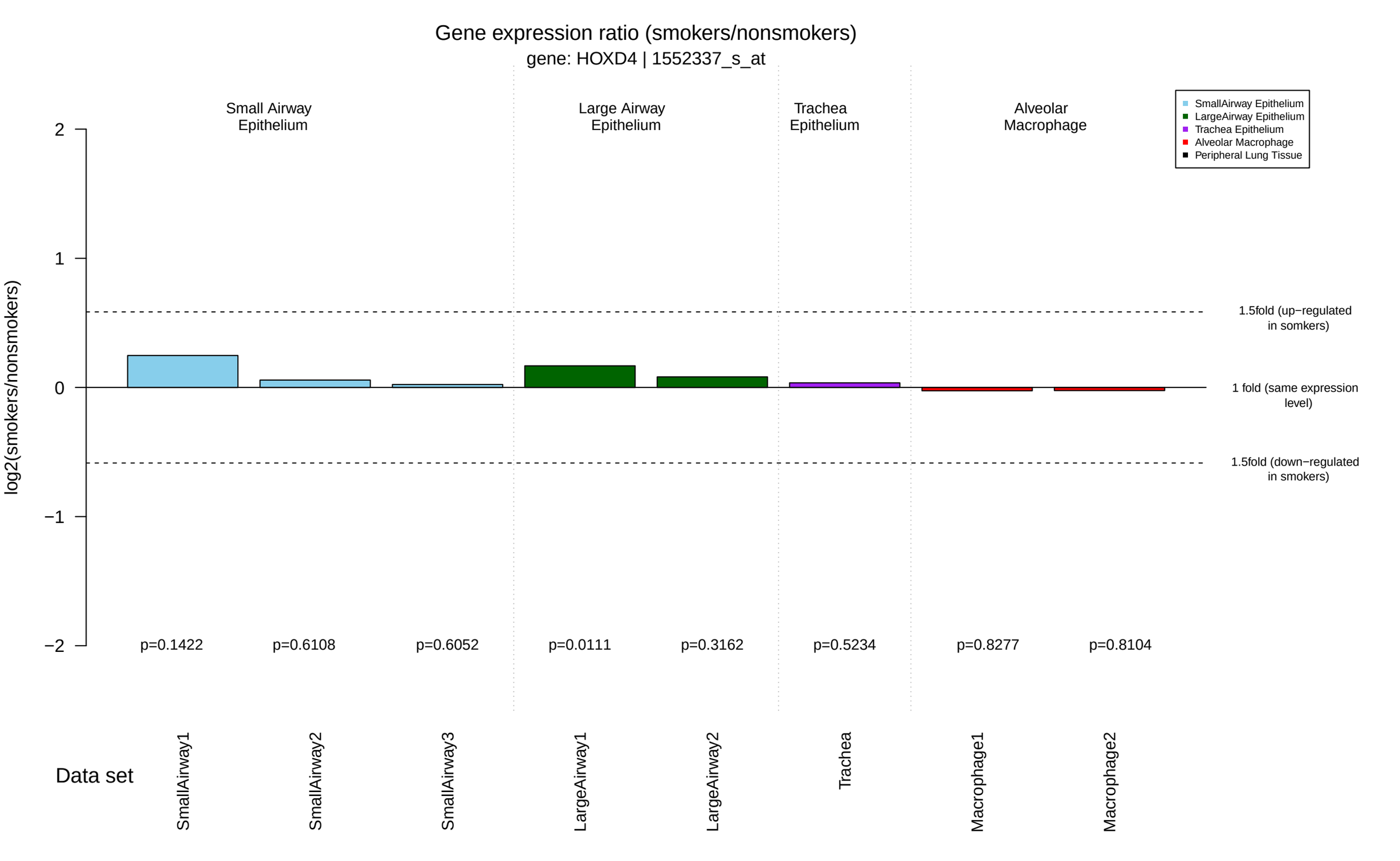
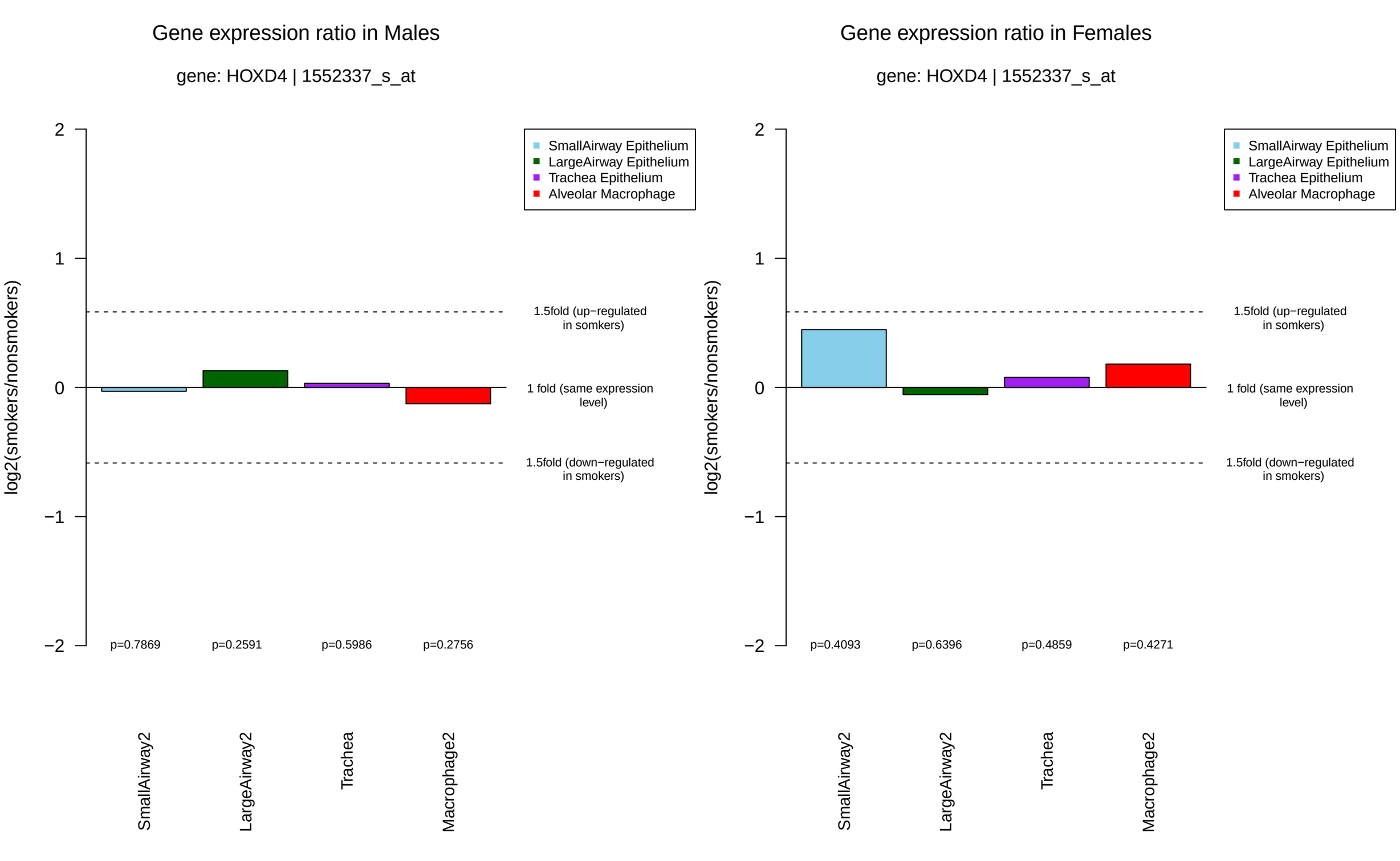
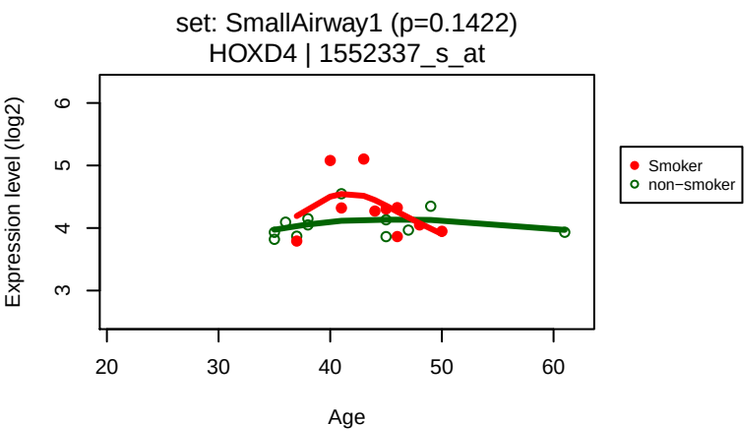
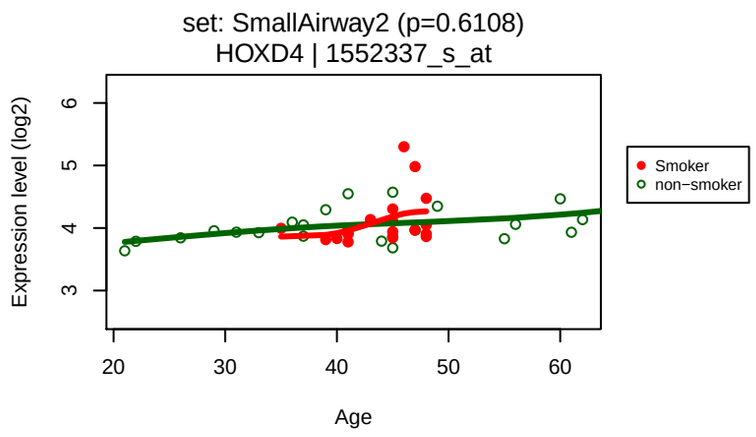
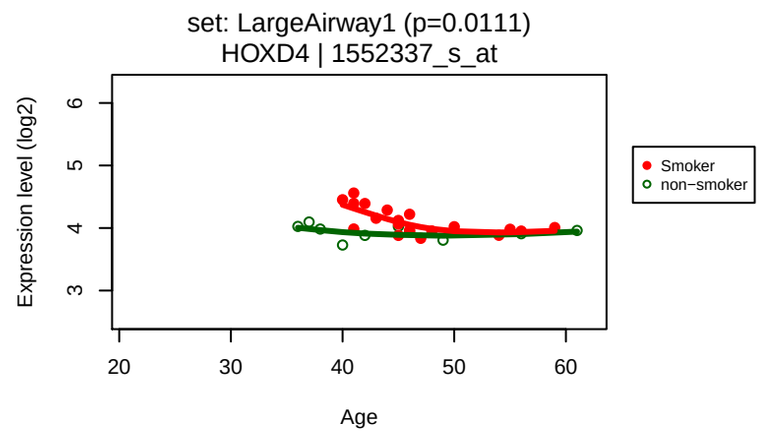

Gene expression of smokers and nonsmokers
gene: HOXD4 | 1552337_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1422
p=0.6108
p=0.6052
p=0.0111
p=0.3162
p=0.5234
p=0.8277
p=0.8104
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

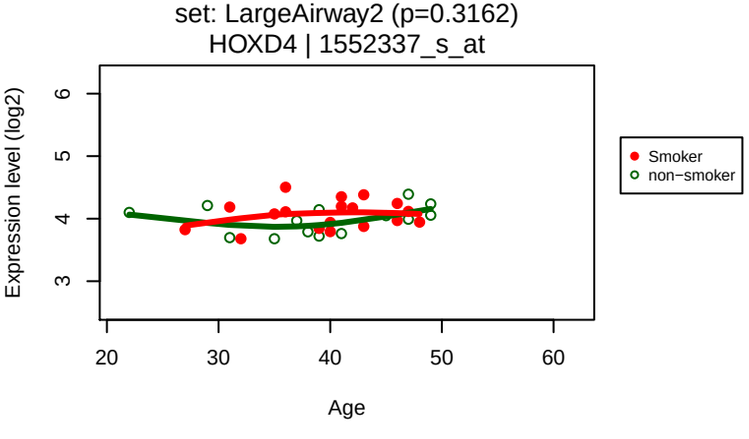

Gene expression ratio (smokers/nonsmokers)
gene: HOXD4 | 1552337_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1422
p=0.6108
p=0.6052
p=0.0111
p=0.3162
p=0.5234
p=0.8277
p=0.8104
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

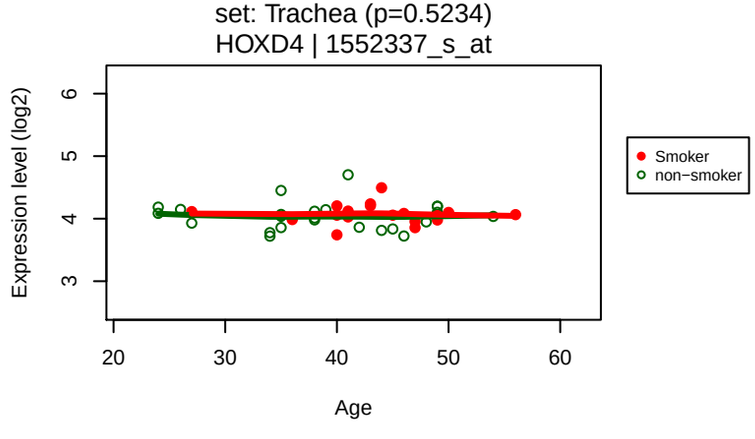

Gene expression ratio in Males
gene: HOXD4 | 1552337_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7869
p=0.2591
p=0.5986
p=0.2756
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXD4 | 1552337_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4093
p=0.6396
p=0.4859
p=0.4271
SmallAirway2
LargeAirway2
Trachea
Macrophage2

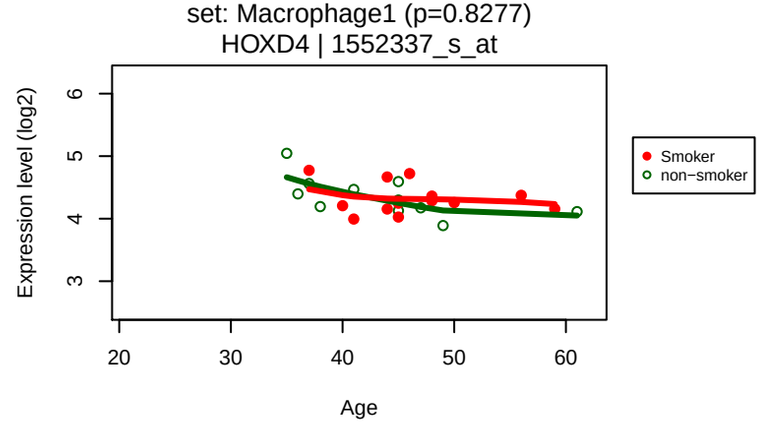

set: SmallAirway1 (p=0.1422)
HOXD4 | 1552337_s_at
Expression level (log2)
Age
Smoker
non−smoker

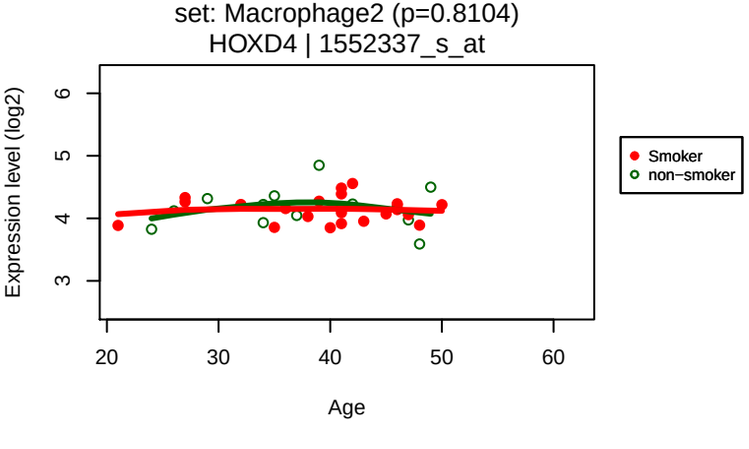

set: SmallAirway2 (p=0.6108)
HOXD4 | 1552337_s_at
Expression level (log2)
Age
Smoker
non−smoker

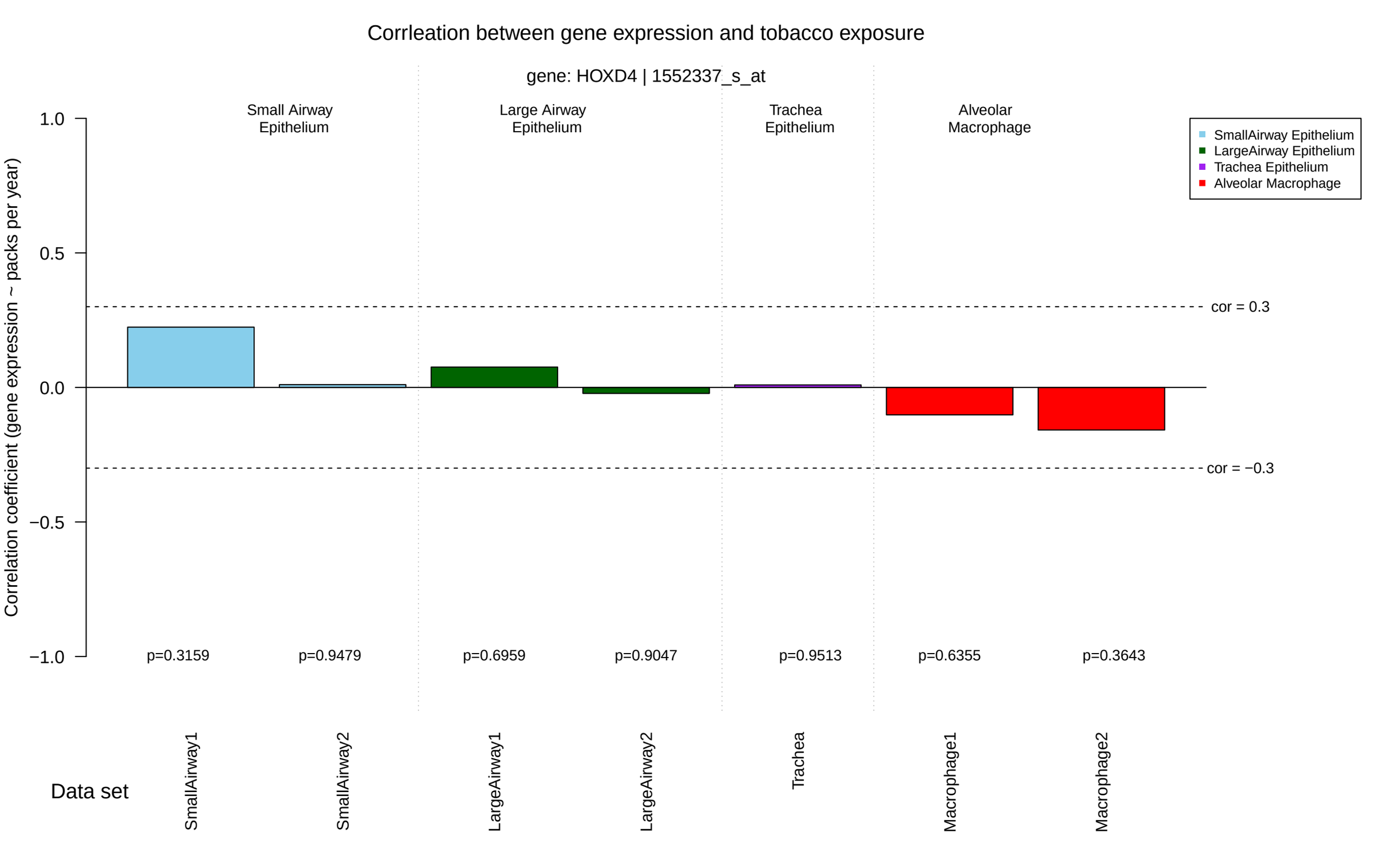

set: LargeAirway1 (p=0.0111)
HOXD4 | 1552337_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3162)
HOXD4 | 1552337_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5234)
HOXD4 | 1552337_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8277)
HOXD4 | 1552337_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8104)
HOXD4 | 1552337_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXD4 | 1552337_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3159
p=0.9479
p=0.6959
p=0.9047
p=0.9513
p=0.6355
p=0.3643
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2